```
ubuntu@ip-172-31-5-35:~/Trend$ wget -O - https://apt.releases.hashicorp.com/gpg | sudo gpg --dearmor -o /usr/share/keyrings/hashicorp-archiv
e-keyring.gpg
echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/hashicorp-archive-keyring.gpg] https://apt.releases.hashicorp.com
 $(grep -oP '(?<=UBUNTU_CODENAME=).*' /etc/os-release || lsb_release -cs) main" | sudo tee /etc/apt/sources.list.d/hashicorp.list
sudo apt update && sudo apt install terraform
```

```
ubuntu@ip-172-31-5-35:~/Trend$ terraform version
Terraform v1.14.3
on linux_amd64
ubuntu@ip-172-31-5-35:~/Trend$
```

```
ubuntu@ip-172-31-5-35:~/Trend$ curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
unzip awscliv2.zip
sudo ./aws/install
```

```
ubuntu@ip-172-31-5-35:~/Trend$ sudo apt install unzip
Reading package lists... Done
Building dependency tree... Done
```

```
ubuntu@ip-172-31-5-35:~/Trend$ unzip awscliv2.zip
```

```
ubuntu@ip-172-31-5-35:~/Trend$ sudo ./aws/install
You can now run: /usr/local/bin/aws --version
ubuntu@ip-172-31-5-35:~/Trend$
```

```
ubuntu@ip-172-31-5-35:~/Trend$ mkdir terraform
ubuntu@ip-172-31-5-35:~/Trend$ cd terraform
ubuntu@ip-172-31-5-35:~/Trend/terraform$
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ touch main.tf variables.tf provider.tf outputs.tf
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ nano provider.tf
```

GNU nano 7.2 provider.tf *

```
provider "aws" {
  region = "ap-south-1"
}
```

```
main.tf  outputs.tf  provider.tf  variables.tf
ubuntu@ip-172-31-5-35:~/Trend/terraform$ nano main.tf
```

```
resource "aws_subnet" "public_subnet" {
  vpc_id                  = aws_vpc.trend_vpc.id
  cidr_block              = "10.0.1.0/24"
  availability_zone       = "ap-south-1a"
  map_public_ip_on_launch = true

  tags = {
    Name = "trend-public-subnet"
  }
}
```

```
resource "aws_internet_gateway" "trend_igw" {
  vpc_id = aws_vpc.trend_vpc.id

  tags = {
    Name = "trend-internet-gateway"
  }
}
```

```
resource "aws_route_table" "public_rt" {
  vpc_id = aws_vpc.trend_vpc.id

  route {
    cidr_block = "0.0.0.0/0"
    gateway_id = aws_internet_gateway.trend_igw.id
  }

  tags = {
    Name = "trend-public-route-table"
  }
}
```

```
resource "aws_route_table_association" "public_assoc" {
  subnet_id      = aws_subnet.public_subnet.id
  route_table_id = aws_route_table.public_rt.id
}
```

```
resource "aws_security_group" "jenkins_sg" {
  name ="jenkins-security-group"
  description = "allow ssh, jenkins and app traffic"
  vpc_id = aws_vpc.trend_vpc.id

  ingress {
    description = "ssh access"
    from_port = 22
    to_port = 22
    protocol = "tcp"
    cidr_blocks = [0.0.0.0/0]
}

  ingress {
    description = "jenkins ui"
    from_port = 8080
    to_port = 8080
    protocol = "tcp"
    cidr_blocks = [0.0.0.0/0]
}

  ingress {
    description = "application port"
    from_port = 3000
    to_port = 3000
    protocol = "tcp"
    cidr_blocks = [0.0.0.0/0]
}

  egress {
    description = "ssh access"
    from_port = 22
    to_port = 22
    protocol = "-1"
    cidr_blocks = [0.0.0.0/0]
}

  tags = {
    Name = "trend-jenkins-sg"
 }
}
```

```
resource "aws_instance" "jenkins_ec2" {
  ami                    = "ami-0f5ee92e2d63afc18"
  instance_type          = "t3.micro"
  subnet_id              = aws_subnet.public_subnet.id
  vpc_security_group_ids = [aws_security_group.jenkins_sg.id]

  tags = {
    Name = "jenkins-server"
  }
}
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/aws...
- Installing hashicorp/aws v6.27.0...
- Installed hashicorp/aws v6.27.0 (signed by HashiCorp)
Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
ubuntu@ip-172-31-5-35:~/Trend/terraform$
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ terraform plan

Planning failed. Terraform encountered an error while generating this plan.


 Error: No valid credential sources found

   with provider["registry.terraform.io/hashicorp/aws"],
   on provider.tf line 1, in provider "aws":
    1: provider "aws" {

 Please see https://registry.terraform.io/providers/hashicorp/aws
 for more information about providing credentials.

 Error: failed to refresh cached credentials, no EC2 IMDS role found, operation error ec2imds: GetMetadata, http response error
 StatusCode: 404, request to EC2 IMDS failed
```

🔍 Ye error kyon aata hai?

Tumhara setup dekh kar lag raha hai:

- Tum **EC2 Ubuntu machine** par Terraform chala rahe ho
- EC2 par **IAM Role attached nahi hai**
- AWS CLI credentials bhi configure nahi kiye

Terraform AWS provider **4 jagah credentials dhundta hai**:

1. Environment variables
2. `~/.aws/credentials` file
3. AWS CLI config
4. EC2 IAM Role (IMDS)

👉 Tumhare case me **koi bhi available nahi hai**, isliye error.

Create role

Trusted entity type

- (selected) **AWS service**
  Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- **AWS account**
  Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- **Web identity**
  Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- **SAML 2.0 federation**
  Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- **Custom trust policy**
  Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case

EC2

Choose a use case for the specified service.

Use case

- (selected) EC2
  Allows EC2 instances to call AWS services on your behalf.

Add permissions Info

Permissions policies (1/1125) Info

Choose one or more policies to attach to your new role.

Filter by Type

amazonec2full | All types | 1 match | 1

	Policy name	Type	Description
☑	AmazonEC2FullAccess	AWS managed	Provides full access to Amazon EC2 via...

Set permissions boundary - *optional*

Cancel Previous Next

Name, review, and create

Role details

Role name

Enter a meaningful name to identify this role.

terraform-ec2-role

Maximum 64 characters. Use alphanumeric and '+=,.@-_' characters.

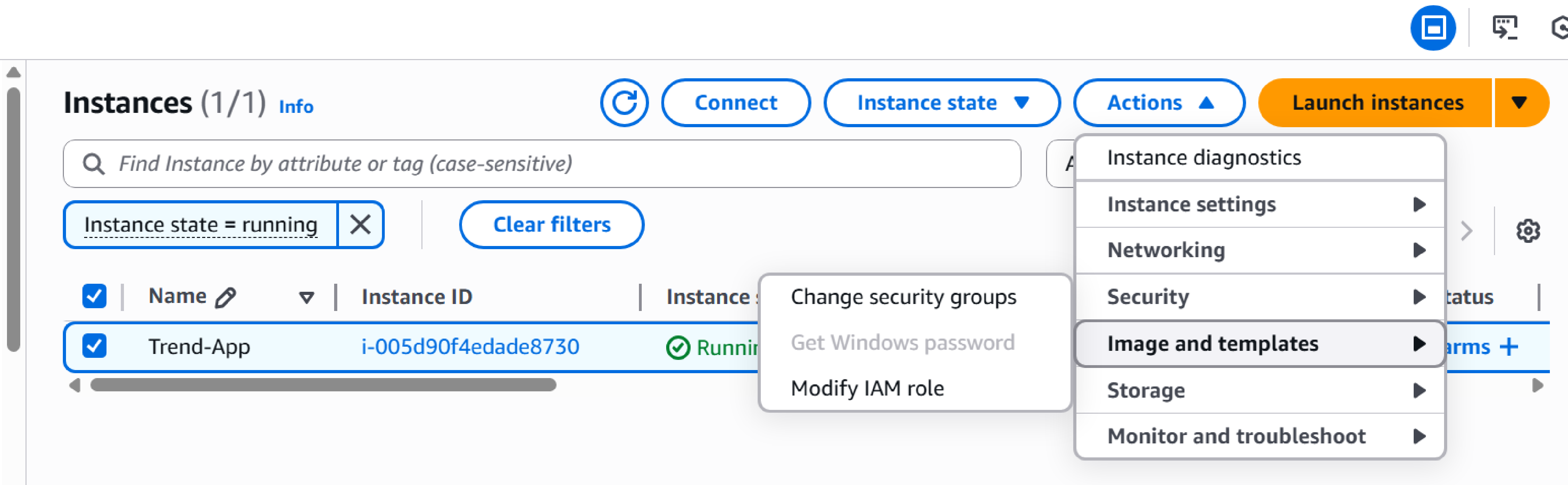
Description

Add a short explanation for this role.

Allows EC2 instances to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _+=,. @-/\[{}]!#$%^*();:"' `

Cancel Previous Create role


Instances (1/1) Info
Connect
Instance state
Actions
Launch instances
Find Instance by attribute or tag (case-sensitive)
Instance state = running
Clear filters
Name
Instance ID
Instance
Trend-App
i-005d90f4edade8730
Runnir
Change security groups
Get Windows password
Modify IAM role
Instance diagnostics
Instance settings
Networking
Security
Image and templates
Storage
Monitor and troubleshoot
tatus
arms

Modify IAM role Info

Attach an IAM role to your instance.

Instance ID

i-005d90f4edade8730 (Trend-App)

IAM role

Select an IAM role to attach to your instance or create a new role if you haven't created any. The role you select replaces any roles that are currently attached to your instance.

terraform-ec2-role

Create new IAM role

Cancel

Update IAM role

```
terraform plan
```

```
Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.jenkins_ec2 will be created
  + resource "aws_instance" "jenkins_ec2" {
      + ami                                  = "ami-0f5ee92e2d63afc18"
      + arn                                  = (known after apply)
      + associate_public_ip_address          = (known after apply)
      + availability_zone                    = (known after apply)
      + disable_api_stop                     = (known after apply)
      + disable_api_termination              = (known after apply)
      + ebs_optimized                        = (known after apply)
      + enable_primary_ipv6                  = (known after apply)
      + force_destroy                        = false
      + get_password_data                    = false
      + host_id                              = (known after apply)
```

```
Plan: 7 to add, 0 to change, 0 to destroy.
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ terraform validate
Success! The configuration is valid.

ubuntu@ip-172-31-5-35:~/Trend/terraform$
```

```
ubuntu@ip-172-31-5-35:~/Trend/terraform$ terraform apply

Terraform used the selected providers to generate the following execution plan. Resource actions are indicated with the following symbols:
  + create

Terraform will perform the following actions:

  # aws_instance.jenkins_ec2 will be created
  + resource "aws_instance" "jenkins_ec2" {
      + ami                                    = "ami-0f5ee92e2d63afc18"
      + arn                                    = (known after apply)
      + associate_public_ip_address            = (known after apply)
```

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_vpc.trend_vpc: Creating...
aws_vpc.trend_vpc: Creation complete after 1s [id=vpc-0f8363ff3debe035c]
aws_internet_gateway.trend_igw: Creating...
aws_subnet.public_subnet: Creating...
aws_security_group.jenkins_sg: Creating...
aws_internet_gateway.trend_igw: Creation complete after 0s [id=igw-0c0ffa5b64714ef89]
aws_route_table.public_rt: Creating...
aws_route_table.public_rt: Creation complete after 1s [id=rtb-0df993a986a80abf1]
aws_security_group.jenkins_sg: Creation complete after 2s [id=sg-008e9fff702b00f62]
```

```
Apply complete! Resources: 7 added, 0 changed, 0 destroyed.
ubuntu@ip-172-31-5-35:~/Trend/terraform$
```

☑	jenkins-server	i-0487941227a86f4db	Running	t3.micro	3/3 checks passec	View alarms +